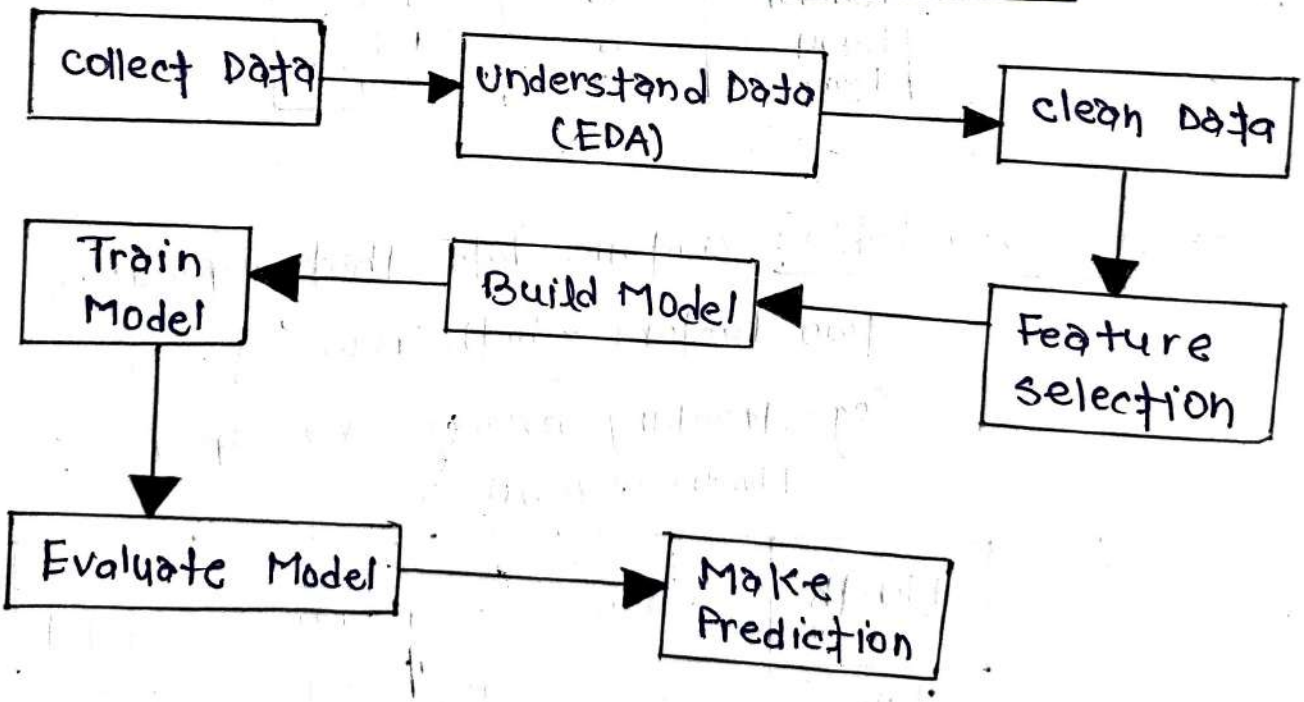
STATISTICS

→ **statistical visualization**: It is the process of representing data graphically to understand its structure, patterns, relationships & anomalies before building Models.

In ML, data visualization is the first step of Exploratory Data Analysis (EDA).

→ **Role of statistics in the ML pipline**



1. STATISTICAL VISUALIZATION

→ **Data Tables** : A Tabular representation of data in rows & columns.

↳ **One-Way Table** : Contains one individual with variable that depend solely on the individual.
(eg - Name, Age, salary)

ex -

Name	Age	salary
Aniket	20	1.5
Ramlal	23	1.6
Raju	25	1.8
Ram	29	2.0

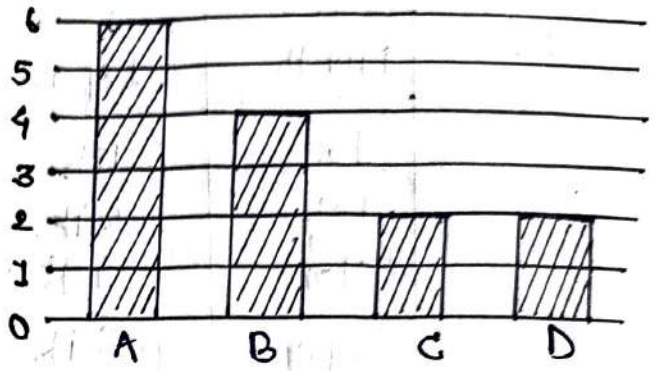
↳ **Two-way Table** : Contains data that depends on two factors - both rows & colums
(eg - tracking revenue over specific Months & years)

ex -

Month	2020	2021	2023	total
Jan	5K	10K	15K	30K
Mar	10K	15K	25K	50K
May	10K	35K	10K	55K
Jul	15K	20K	25K	60 K
Sept	5K	5K	20K	30K

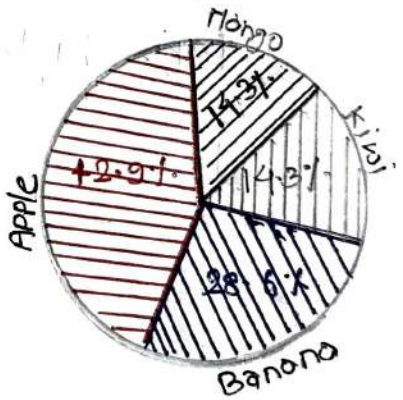
→ **Bar graph** : visualizes the frequency (count) of categorical variables by using horizontal or vertical bars.

Persons	Fruits
A	6
B	4
C	2
D	2

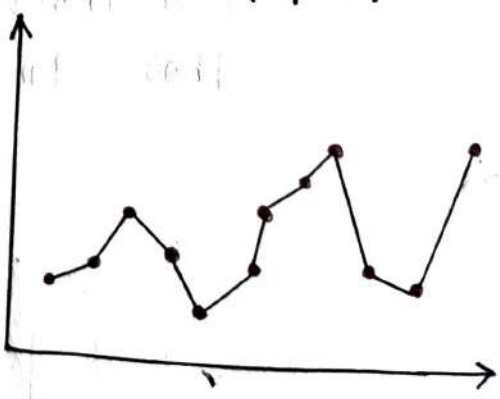


→ **Pie chart** : Visualizes data based on the percentage occurance of variables, slicing & circle to represent parts of whole (100%)

fruits	count	
Apple	6	→ 42.9%
Banana	4	→ 28.6%
kiwi	2	→ 14.3%
Mango	2	→ 14.3%



→ **Line Graph** : created by placing dots on top of frequency bars & connecting them. It is highly useful for visualizing the flow or trend of continuous data overtime.

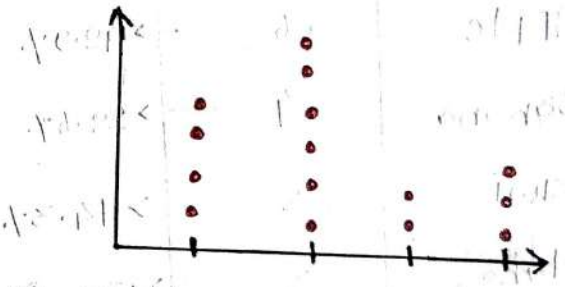


→ **O-Gives** : A specific type of line graph created by plotting cumulative frequencies (adding up frequencies as go) ↳ adding after and after.

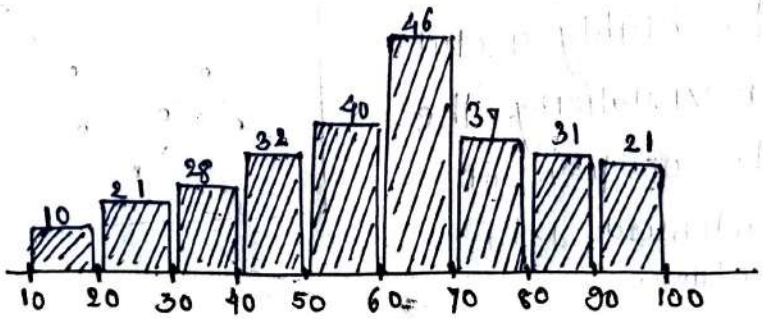
Or

Month	Expense
Jan	10K. ↓ adding
Feb	24K ↓ adding
Mar	42K ↓ adding
Apr	54K ↓ adding
May	59K

→ **Dot Plot**: Build by stacking dots to represent frequency. It builds a foundational understanding for histogram.



→ **Histogram**: Groups numeric data into specific buckets or "bins" (eg., 10-20, 20-30). The bars represent the Number of element falling inside these bins.



2. MEASURE OF CENTRAL TENDENCY

↳ Mathematical techniques used to describe the exact middle point of our data, acting as a crucial foundation for ML algorithms.

→ **Mean (Average)** : calculated by summing all numbers & dividing by total count.

It acts as a "balancing point" on a like seesaw.

$$\text{Mean} = \frac{\text{sum of all NO.}}{\text{total count}}$$

• Fancy representation:-

$$\text{Mean} = (\bar{x}) = \frac{1}{N} \sum_{i=1}^{N} x_i$$

N → Nth last count
Σ → summation
x_i → ith dat value (x_1, x_2, x_3 etc)
$\frac{1}{N}$ → counts
$i=1$ → counter (Like - 1st, 2nd, 3rd)

ex- Data = 2, 2, 7, 9, 10

$N=5$, $x_1 = 2$, $x_2 = 2$, $x_3 = 7$, $x_4 = 9$, $x_5 = 10$

$$\bar{x} = \frac{1}{5} \sum_{i=1}^{5} x_i$$

$$= \frac{1}{5}(2+2+7+9+10) = \frac{30}{5} = 6$$

→ **Median** : The exact midpoint of our data when arranged in order. For an odd number of data points, it is the Middle number.

for even number of data points, we find the avg of the two middle numbers.

→ **Mode** : The number that occurs most frequently in our data.

If all numbers are distinct, there is no mode.

If two or more numbers tie for the highest frequency, the dataset is "bimodal" or "Multimodal".

eg- data = 1, 2, 1, 1, 2, 2, 4, 8, 9

2 & 1 ; both are two diff modes

∴ thus bimodal.

3. MEASURE OF SPREAD/DISPERSION

↳ It Describes how tightly clustered or widely scattered our data is around that center point.

→ **Range**: tell us the distribution of data by subtracting the smallest data point from the largest data point.

$$\text{Range} = X_{max} - X_{min}$$

X_{max}: Max value, X_{min}: Min value

→ **IQR (Interquartile Range)**: Measures the distance between different quarters (percentiles) of our data. The data is divided into

Quaters:

Q1 - 25 percentile
Q2 - 50 percentile
Q3 - 75 percentile
Q4 - 100 percentile

$$IQR = Q_3 - Q_1$$

IQR is calculated by subtracting Q1 from Q3 (IQR = Q3 - Q1)
If the IQR is low, the data is very close to the median.

ex

12, 12, 13, 14, [15], 17, 20, 20, 28 — 29, 31, 35, 46, [48], 49, 51, 59, 60

Q1 Q2 Q3

4. OUTLIERS

↳ Outliers are extreme, abnormally large, or abnormally small values in our dataset.
(ex- a value of 1001 in dataset of sigal digit no.)

→ **The Impact of outlier's**

- It can distort measures like mean & standard deviation, making them less representative of the data
- In ML, especially regression models, outliers can disproportionately influence the Model, leading to poor generalization.
- outlier impact our model performance our statistical testing & many other things so we have to get rid from them.

→ **Identification & Removal**: outlier are removed by establishing an Upper fence & a Lower-fence using IQR.

- Upper Fence: $Q3 + 1.5 \times IQR$
- Lower Fence: $Q1 - 1.5 \times IQR$

any value that falls above the upper fence or below the lower fence is considered an outlier and is discarded.

ex- 1, 1, 2, 2, 3, 4, 5, 5, 6, 6, 7, 8, 9, 9, 10, 10, 11, 13, 13, 14, 15, 101

sol-

1, 1, 2, 2, 3, 4, 5, 5, 6, 6, 7, 8, 9, 9, 10, 10, 11, 13, 13, 14, 15, 101

↓ Q1 ↓ Q2 ↓ Q3

$IQR = Q_3 - Q_1 = 7$

$= 11 - 4$

$Q_1 = 4$

$Q_3 = 11$

Now

- Upper Fence $= Q_3 + 1.5\,(IQR)$

$= 11 + 1.5(7) = 21.5$

- Lower fence $= Q_1 - 1.5\,(IQR)$

$= 4 - 1.5(7) = -6.5$

That means value above 21.5 & value below -6.5 are "outliners". so in our data 101 is the only element that is outliner.

5. FIVE NUMBER SUMMARY

The "five-number summary" is a statistical measure used to describe the distribution of a dataset.

It provide a quick overview of the data's spread & center & it includes the following five key values:

- Minimum
- First Quartile (Q1)
- Median (Q2)
- Third Quartile (Q3)
- Maximum

It is crucial that outliers are completly removed from the dataset before calculating these five numbers.

ex - 2, 4, 7, 10, 15, 25, 25 outlier (Upper Fence = $Q_3 + 1.5 \times IQR$ = $20 + 1.5 \times (Q_3 - Q_1)$ = 44)

- Minimum = 2
- Q1 = 4
- Median (Q2) = 10
- Q3 = 20
- Maximum = 25

2 4 7 10 15 20 25

Q1 Q2 Q3

6. OUTLIERS CODE

This code demonstrate how outliers can be detected & removed from a dataset in Python using Numpy & the IQR technique. It filters the data by keeping only the values within the lower & upper limits & removes extreme value such as outliers.

In [1]:

```
import numpy as np
arr = np.array ([2, 5, 4, 6, 7, 8, 9 , 12, 13, 16, 17,
                 23, 25, 27, 34, 37 , 201 ])
arr
```

In [2]:

```
Q1 = np.percentile (arr, 25 )
Q3 = np.percentile (arr, 75)
#For Finding Percentile
```

In [3]:

```
IQR = Q3 - Q1      # for finding IQR
IQR
```

output = 18.0

In [4]:

```
UF = Q3 + (1.5 * IQR)
LF = Q1 - (1.5 * IQR)
```

#calculating Fence : lower & Upper

In [5]:

```
l = [ ]
for i in arr:
    if i >= LF and i <= UF
        l.append(i)
arr2 = np.array(l)
```

We create an empty list, loop through the original array (for i in arr:), and conditionally append elements only if they fall strictly inside the fence (if i >= lower_fence and i <= upper_fence).

In [6]

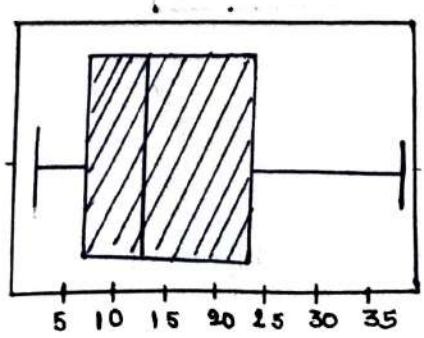
```
print (arr2)
```

output: [2 3 4 6 7 8 9 12 13 16 17 23 25
27 34 37]

In [7]

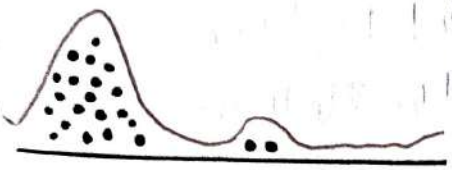
```
import seaborn as sns
sns.boxplot (x = arr2)
```

output:



DATA DISTRIBUTION

↳ It Describe how data values are spread, clustered & arranged around a central value (usually the mean) It helps us understand the pattern of the data, such as whether the data is concentrated, widely dispersed, symmetrical or skewed.

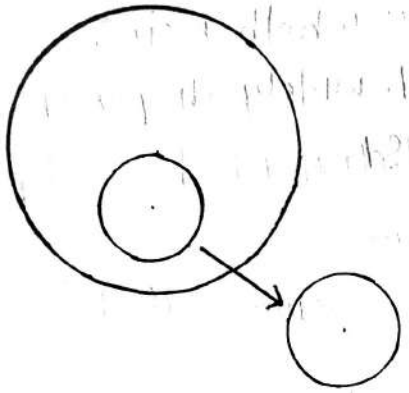


→ Role of Data Distribution in ML

- Helps understand the pattern & spread of data
- Detects outliers & anomalies.
- Helps choose appropriate ML algorithms & preprocessing techniques.
- Improves Model performance & prediction accuracy.

7. VARIANCE & STANDARD DEVIATION

* **Population**: The entire collection of individuals objects or data points that we want to study
like- world population in total is 8 Billion.
ex- whole circle is population



* **sample**: A smaller subset of the population selected to represent the entire population & draw conclusion about it.
like- 500 people selected for testing a vaccine
ex- small circle cut from whole circle.

→ **Variance** (σ^2) : It is a statistical measure that quantifies the spread or dispresion of a set of data point around their mean (average)

Fancy representation formula

$$\sigma^2 = \frac{1}{N} \sum_{j=1}^{N} (x_j - \mu)^2$$

σ^2 ← Variance
N ← Total No. of Data point
$\sum$ ← summation
x_j ← individual Data point
μ ← Population (Mean)

ex - 85, 90, 95, 100, 105

mean = 95

Now

$(85-95)^2 + (90-95)^2 + (95-95)^2 + (100-95)^2 + (105-95)^2 = 250$

$\frac{250}{5} = 50$ ✓

→ **Standard deviation**: It represents the average distance that a single data point sits away from the mean.

Found by taking exact square root of the variance

ex- variance = 50

Standard Deviation $= \sqrt{50} = 7.07$

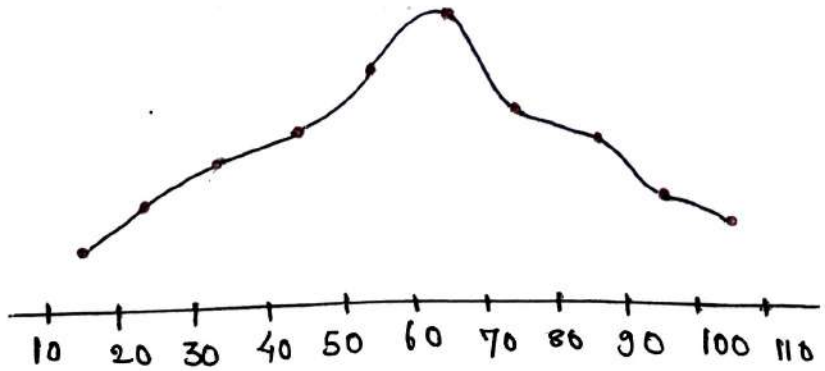
Formula

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

- σ (sigma) = standard deviation
- N = No. of Data point
- Σ = Summation
- i = counter
- x_i = individual Data point
- μ = population Mean

→ **Density curve**

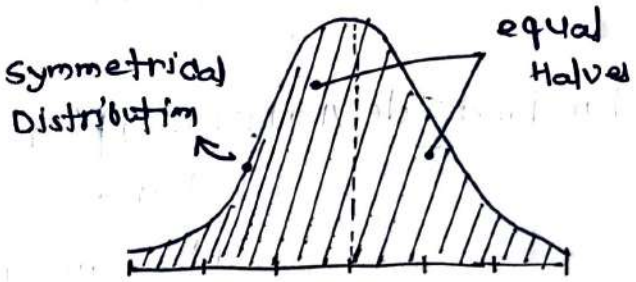
It illustrates the entire distribution of our data. The Total area underneath any density curve is exactly 100% (or 1).



* higher the bins higher the accuracy lower the bins lower the accuracy.

→ **Symmetrical Distribution**

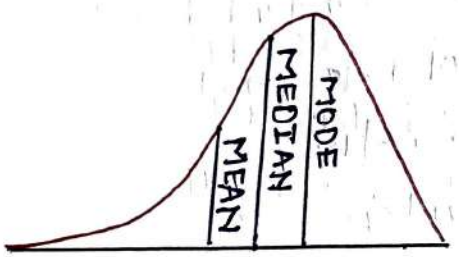
It is also called as a "normal distribution" & it always follows a bell shaped curved.

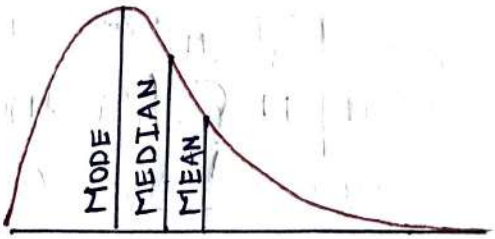


Mean = Median = Mode

→ **Skewed Distribution**

Occurs when extreme outliers pull the tail of the curve either to the left or right.





In skewed curve, the Mean is severely pulled towards the tail, the Mode stays at the highest density peak, & the Median stably remains in the center.

8. Z-SCORE

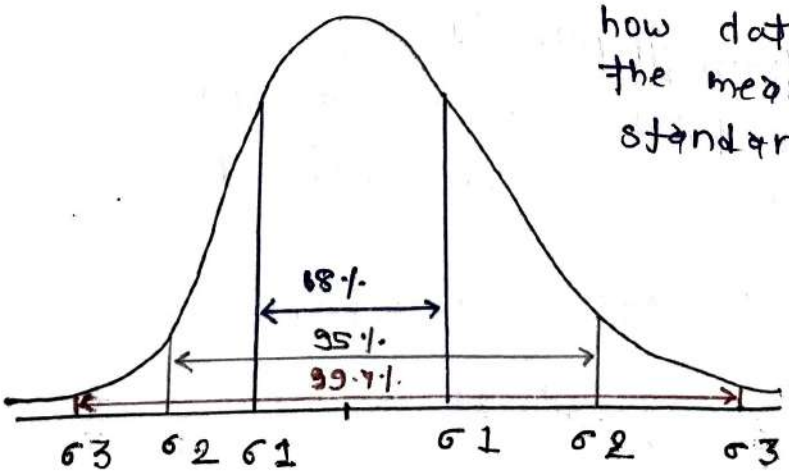
↳ It is a powerful statistical tool used to determine exactly how far a specific data point is from the mean, specifically measured in "standard deviation points".

- **Formula:**

$$z = (\text{Data points} - \text{Mean}) / \text{standard deviation}$$

↳ A positive z score indicates the data point lies to the right side of the mean, whereas a -ve z score indicates it lies to the left.

→ **68-95-99.7 Rule (Empirical Rule):** It applies to a normal (bell shaped) distribution & describe how data is distributed around the mean (μ) in terms of standard deviation (σ).



- 68% of data lies within ± 1 standard deviation (σ) from the mean (μ).
- 95% of data lies within $\pm 2\sigma$ from the mean.
- 99.7% of data lies within $\pm 3\sigma$ from the mean.

* A Normal distribution never reaches 100% within $\pm 3\sigma$ because its tails extend infinitely, leaving about 0.3% of data outside this range.

→ **z-Table Usage** :- once a z score is calculated, we can look up that exact value on a z Table to find the specific percentage of data that falls below (or above) that data point within a normal distribution

ex- suppose mean = 8, standard deviation is 1 & Standard deviation point away is 9.5 from the mean. (data point = 9.5) find z score.

Z score = $(9.5 - 8)/1 = 1.5$

Positive z score.

9. BASIC PROBABILITY

→ **Probability** : chances of an event occuring, always resulting in a Mathematical value perfectly between 0 & 1 (which can also be multiplied by 100 to represent 0% to 100%.)

Formula = True outcomes / Total outcomes

eg - If we flip a coin, the True outcome = 1
Total outcomes = 2 (Head or Tail)

Probability = 1/2

→ **Role of Probability in Machine learning (ML)**

- Helps predict the likelihood of outcomes.
- Enables decision-making under uncertainty
- Forms the basis of many ML algorithms & probabilistic Models.
- Helps estimate the confindence of predictions.

In one line :- "It allows machine learning Models to make pridictions & decisions in uncertain situations".

14. PROBABILITY EVENTS

→ **Independent Events** : occurance of one event does not affect the probability of another event.

condition:

$$P(B/A) = P(B)$$

Formula:

$$P(A \cap B) = P(A) \times P(B)$$

↳ Intersection symbol

- $P(A)$ = Probability of event A
- $P(B)$ = Probability of event B
- $P(A \cap B)$ = Probability of both events occuring

ex- Tossing a coin twice

$P(A) = 1/2$ (Head or Tail)
$P(B) = 1/2$

$P(A \cap B) = P(A) \times P(B)$
$= 1/2 \times 1/2 = 1/4$

→ **Dependent Events** : occurance of one event affects the probability of another event.

Formula:

$$P(A \cap B) = P(A) \times P(B/A)$$

- $P(B/A)$ = Probability of B given A.

ex- A standard deck contains 52 playing cards, including 4 Kings. Two cards are drawn one after another without replacement. What is the probability that both cards drawn are kings?

- $P(A) = \frac{4}{52}$ { Probability of getting the first king }

- $P\left(\frac{B}{A}\right) = \frac{3}{51}$ { Probability of getting the 2nd kind, given that the first was king.

$$P(A \cap B) = P(A) \times P(B/A)$$

or

$$P(\text{Both kings}) = P(A) \times P(B/A)$$

$$= \frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$$ ≡

→ Mutually Exclusive Events : Two events cannot occur at the same time

Condition:

$$P(A \cap B) = 0$$

Formula:

$$P(A \cup B) = P(A) + P(B)$$

↳ Probability of the Union of events A & B.

- $P(A \cup B)$ = probability that either event A or event B (or both) occurs.

ex- Getting 1 & 6 on a single die roll.

$P(A) = 1/6$, $P(B) = 1/6$

$$P(A \cup B) = P(A) + P(B)$$

$$= \frac{1}{6} + \frac{1}{6} = \frac{2}{6}.$$

$$= 1/3$$

12. ADDITION & MULTIPLICATION RULE

→ **Addition Rule (OR/Union)**: used to find the probability at least one event occurs.

General Formula:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- $P(A \cup B)$ = Probability of the union of event A & B.
- $P(A \cap B)$ = Probability of Both events occuring.

ex:- Drawing one card from a deck.

↳ P(A): getting a Heart (13/52)

P(B): getting a King (4/52)

$P(A \cap B)$: King of Heard (1/52)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{13}{52} + \frac{4}{52} - \frac{1}{52}$$

$$= \frac{16}{52} = 4/13$$

- **For Mutually Exclusive events**

$$P(A \cup B) = P(A) + P(B)$$

→ Multiplication Rule (AND / intersection) used to find the probability of both events occuring together

For independent event

$$P(A \cap B) = P(A) \times P(B)$$

For Dependent Events

$$P(A \cap B) = P(A) \times P(B/A)$$, where $P(B/A) = \frac{P(A \cap B)}{P(A)}$

ex- Drawing two kings from a deck without replacement

↳ A: first card is a king $(4/52)$

B/A: second card is a king, given the first was a king $(3/51)$

$$P(A \cap B) = \frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$$

13. CONDITIONAL PROBABILITY

↳ Probability of event A occurring given that event B has already occured.

mean probability of an event happening when another event has already happend.

Formula:

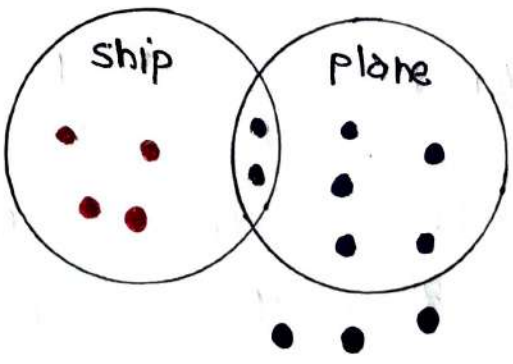
$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

- $P(A/B)$ = probability of A given B
- $P(A \cap B)$ = joint probability A & B
- $P(B) \neq 0$

ex- A survey of 14 students was conducted about their preferences for ships & planes. The result are shown in the table below.

↳ let make a table

	love ship	no ship
love Plane	2	5
No Plane	4	3



Now solution

(1) $P(\text{Loves ship} / \text{Love plane}) = \frac{P(\text{Loves ship} \cap \text{Love plane})}{P(\text{Love Plane})}$

$= \frac{2/14}{7/14} = \frac{2}{14} \times \frac{14}{7} = \frac{2}{7}$

(2) $P(\text{Love Plane} / \text{Loves ship}) = \frac{P(\text{Loves Plane} \cap \text{Loves ship})}{P(\text{Loves ship})}$

$= \frac{2/14}{6/14} = \frac{2}{14} \times \frac{14}{6} = \frac{2}{6} = 1/3$

(3) $P(\text{Love Plane} / \text{No ship}) = \frac{P(\text{Love Plane} \cap \text{No ship})}{P(\text{No ship})}$

$= \frac{5/14}{8/14} = \frac{5}{14} \times \frac{14}{8} = \frac{5}{8}$

(4) $P(\text{Loves ship} / \text{No plane}) = \frac{P(\text{Loves ship} \cap \text{No plane})}{P(\text{No plane})}$

$= \frac{4/14}{7/14} = \frac{4}{14} \times \frac{14}{7} = \frac{4}{7}$

Now the results -

- $P(\text{Loves ship} / \text{Love plane}) = 2/7$
- $P(\text{Loves plane} / \text{Loves ship}) = 1/3$
- $P(\text{Loves plane} / \text{No ship}) = 5/8$
- $P(\text{Loves ship} / \text{No plane}) = 4/7$

14. BAYE'S THEOREM

↳ Used to update the probability of an event when new information or evidence is available.

ex-

Old belief + New evidence = Updated belief

intial belief: Aniket thinks his brother chotu has 5 Mangos

New evidence: The next day, aniket sees Chotu his brother has 10 Mangoes.

Belief updated: Aniket changes his belief & now thinks "chotu has 10 Mangoes, not 5"

Formula:

$$P(A/B) = \frac{P(B/A) \times P(A)}{P(B)}$$

- P(A) = Prior Probability
- P(B/A) = Likelihood
- P(B) = Evidence
- P(A/B) = Posterior Probability

Baye's Theorem is heavily utilized in Machine learning (ML), specifically forming the mathmatical backbone of the Naive Bayes algorithm.

Que- A person initially believes there is a 20% chance of rain ($P(A) = 0.2$). If dark cloud are seen (B), the probability of seeing dark clouds when it rains is 80% ($P(B/A) = 0.8$), and the probability of seeing dark clouds is 40% ($P(B) = 0.4$). find updated probability of rain.

↳ $$P(A/B) = \frac{P(B/A) \times P(A)}{P(B)}$$

$$P(A/B) = \frac{0.8 \times 0.2}{0.4} = \frac{0.16}{0.4} = 0.4$$

$$P(A/B) = 0.4 = 40\%$$

15. HYPOTHESIS TESTING BASICS

↳ It is a statistical method to decide whether there is enough evidence in sample confindently support a claim about the whole population.

- Hypothesis testing belongs to "interential statistics" which goes beyond just describing sample data and instead attempts to make predictions or draw conclusion about an entire population.

→ The Hypotheses -

- Null Hypotheses (H_0): Assume there is no difference, change or effect

ex- the coin is fair, not any problem in coin.

It operates on the principle of "innocent until proven guilty".

- $H_0 : \mu = \mu_0$
- Alternate hypothesis (H_1) : The direct opposite of the null hypothesis, representing the claim we are testing for.

ex- the coin is unfair.

- Two tailed $H_1 : \mu > \mu_0$

- Right Tailed $H_1 : \mu > \mu_0$
- Left Tailed $H_1 : \mu < \mu_0$

μ = Actual population (true average of the population)

μ_0 = hypothesized mean (the claimed or assumed average)

→ Significance Value (α) : The probability of rejecting the null hypothesis when it is actualy true. Usually $\alpha = 0.05$ (5%)

→ Confidence Interval (CI): The range in which we expect the true value to lie with a certain confidence.
for $\alpha = 0.05$, the confidence level is 95%.

confidence level $= 1 - \alpha$

Note:- If our experiment's result falls inside this 95%-zone, we accept the "Null hypothesis".
If it falls outside into the 5% margin, we reject the Null hypothesis.

→ **Errors in Testing :**

- **Type 1 Error :** Rejecting the Null hypothesis when it is actually true in reality.

- **Type 2 Error :** Accepting the Null hypothesis when it is actually false in reality.

→ **One-Tailed vs Two-Tailed Tests**

One Tailed Test	Two-Tailed Test
(1) checks if the value is greater than or less than a specific value	(1) check if the value is simply different from a specific value
(2) one direction only (left or right)	(2) Both direction (higher or lower)
(3) significance level (α) = 0.05 or 5% is on one side of the curve	(3) 5% is split into 2.5% on each side
(4) $H_1 : \mu > 70$ or $H_1 : \mu < 70$	(4) $H_1 : \mu \neq 70$

16. Z-TEST

→ Purpose

Test whether a sample mean is significantly different from a population mean.

→ conditions

It Must be used when the population standard deviation (σ) is known, and the sample size is generally greater than 30 ($n > 30$)

→ Formula:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

- $\bar{x}$ = sample mean
- μ = population mean
- σ = population standard deviation
- n = sample size

→ Manual Testing

We calculate the z-score & compare it against a critical value found on a z table ex:- -1.96 & 1.96 for a two-tailed test with a 0.05 significance level.

If the z score falls outside these boundaries, we reject the null hypothesis.

→ Python Implementation

In [1]:

```
import numpy as np
from scipy.stats import norm
```

→ for Mathematical operations like - mean, avg, root etc

→ for statistical function & probability distribution ex - Normal, t, chi-square etc.

In [2]

```
sample = [172, 174, 168, 169, 171, 173, 175, 170, 169, 172]
population_mean = 170
population_std = 3
samp_mean = np.mean(sample)
n = len(sample)
```

In [3]

```
z_score = (samp_mean - population_mean) /
          (population_std / np.sqrt(n))
z_score
```

out [3]:

```
1.3703203194066309
```

In [4]

```
p_value = 2 * (1 - norm.cdf(abs(z_score)))
p_value
```

out [4]:

```
0.17058093284142756
```

In [5]:

```
alpha = 0.05
if p_value < alpha:
    print("I will reject the Null hypothesis")
else:
    print("I will accept the Null hypothesis")
```

out [5]:

I will accept the Null hypothesis.

17. T TEST

→ Purpose

Tests whether a sample mean different significantly from a population mean when the population standard deviation is unknown.

→ Conditions

- We Must use a T-test when the "population standard deviation" is unknown, requiring us to use the sample's standard deviation (s) instead.
- Sample size is small ($n < 30$) or even large when σ is unavailable.

→ Formula:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

- $\bar{x}$ = sample mean
- μ = population mean
- s = sample standard deviation
- n = sample size

→ Degree of Freedom (df): Because we are working with samples, we must calculate the degree of freedom, which is exactly $n-1$.

$$df = n - 1$$

→ Python Implementation

In [1]:

```
import numpy as np
from scipy import stats
```

In [2]:

```
sample = [172, 174, 168, 169, 171, 173, 175, 170, 169, 172]
sample_mean = np.mean(sample)
sample_std = np.std(sample, ddof = 1)
n = len(sample)
mean_population = 170
```

In [3]:

```
t-stats = (sample_mean - mean_population) /
          (sample_std / np.sqrt(n))
t-stats
```

out [3]

```
1.7782460350914434
```

abs removes + & - sign & give only magnitude

In [4]:

```
P-value = 2 * (1 - stats.t.cdf(abs(t-stats), df = n-1))
alpha = 0.05
```

In [5]:

```
if P-value < alpha:
    print ("reject the null hypothesis")
else:
    print ("accept the null hypothesis")
```

out [5]:

```
accept the null hypothesis
```

18. TWO SAMPLE TEST

↳ Two sample tests compare the means of two entirely different groups (ex- comparing test scores between School A & school B)

to see if one performed significantly differently than the other.

In simple words -
compares the means of two independent groups.

(a) Two-sample z-Test

↳ Used when:

- σ_1 & σ_2 are known.
- samples are large

Formula

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2}{n_2}}}$$

(b) Two-sample T-Test

↳ Used when:

- population standard deviations are unknown.
- sample standard deviations are used

Formula (Welch's T-Test)

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- $\bar{x}_1, \bar{x}_2$ = sample means
- s_1, s_2 = sample standard deviations
- n_1, n_2 = sample sizes

→ Degree of Freedom (df)

For two samples, the degree of freedom is calculated by finding the minimum between the two groups:

$$df = \min(n_1 - 1, n_2 - 1)$$

- for equal variances:

$$df = n_1 + n_2 - 2$$

* Why equal_var = False ?

↳ This specific parameter forces python to perform Welch's T-Test, which is the preferred & safer method because it does not assume that both group share equal variances.

→ Python Implementation

In [1]:

```
import numpy as np
from scipy import stats
```

In [2]:

```
group_A = [85,88,90,92,87,85,89,91,86,88]
group_B = [82,84,80,83,81,79,78,85,84,83]
```

In [3]

```
t_stats, p_value = stats.ttest_ind(group_A, group_B,
                                   equal_var = False)
```

In [4]

```
t_stats
```

out [4]

```
5.829604000950716
```

In [5]

```
alpha = 0.05
```

In [6]

```
if p_value < alpha:
    print ("I will reject the null hypothesis")
else
    print("I will accept the null hypothesis")
```

out [6]:

```
I will reject the null hypothesis
```

19. CHI-SQUARE TEST

↳ It is a statistical method used specifically to check if there is a significant association bet^n two categorical variables.

In simple words - checks the relationship between two categorical variables.

* Observed frequency (O_i) :- the actual data collected in a survey or experiment.

* Expected frequency (E_i) :- the Mathematically projected data assumming the null hypothesis is true.

* Expected Frequencies	$= \frac{\text{Row total} \times \text{column Total}}{\text{Grand total}}$

↳ Formula:

$$X^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

- O_i = Observed frequency
- E_i = Expected frequency

→ Degree of freedom (df)

It means the number of values that are free to vary after some constraints are applied.

It calculated differently than other tests, using rows & columns:

$$df = (Rows - 1) \times (column - 1)$$

→ Decision Rule:

- If p-value (in code alpha) < 0.05 → Reject
- If p-value (in code alpha) ≥ 0.05 → Accept

→ Phython Implementation

In[1]:

```
import numpy as np
import seaborn as sns
from scipy.stats import chi_contingency
import pandas as pd
```

→ for numerical operations.

→ for data visualization & creating statistical plots.

→ used to perform the chi-square test of independence.

a library module that provides statistical distribution, hypothesis tests & statistical calculation

→ for data manupulation & analysis using DataFrame & Series

In[2]

```
df = sns.load_dataset('titanic')
df
```

→ To load 'titanic' dataset

In[3]:

```
contigency_table = pd.crosstab(df['sex'], df['survived'])
contigency_table
```

out[3]:

survived	0	1
sex		
female	81	233
male	468	109

Out [8]:

20. ANNOVA TEST

We reject the null hypothesis & there is a significant relationship between gender and survival

20. ANNOVA TEST

↳ It is used to compare the means of more than two independent groups.

Ex- testing if three different teaching Methods (Group A, Group B, Group C) result in significantly different average test score.

It is also known as in mathematically refered as "F-test".

→ Hypotheses

- Null Hypothesis (H_0) = There is no connection or difference between groups.

  All group means are equal.

- Alternate Hypothesis (H_1) = There is a significant difference between at least one of the groups.

→ Decision rule

- If p-value < 0.05 → Reject H_0

→ Python Implementation

In[1]:

```
import seaborn as sns
import pandas as pd
from scipy.stats import f_oneway
```

Statistical Module of the sci-py library

function used to perform one-way ANOVA (analysis of variance) test

In[2]:

```
df = sns.load_dataset('titanic')
df
```

In[3]:

```
df = df[['age','pclass']].dropna()
```

In[4]:

```
df['pclass'].unique()
```

out[4]:

```
array([3,1,2])
```

In[5]:

```
class_1 = df[df['pclass']==1]['age']
class_2 = df[df['pclass']==2]['age']
class_3 = df[df['pclass']==3]['age']
class_1
```

Out [6]:

```
1      38.0
3      35.0
6      54.0
11     58.0
23     28.0
       ...
871    44.0
872    33.0
879    56.0
887    19.0
889    26.0
Name: age, Length: 186, dtype: float64
```

In[7]

```
f_stats, p_value = f_oneway(class_1, class_2, class_3)
```

In[8]

```
f_stats
```

Out [8]

```
57.443484340676214
```

In [9]

```
p_value
```

Out [9]

```
9.489984191956499e-24
```

In [10]:

```
alpha = 0.05
```

In [11]:

```
if P-value < alpha:
    print ("reject the null hypothesis and there is
            a signafacant difference in atleast
            one passenger class")
else:
    print (" there is no signiticant difference")
```

out [11]:

```
reject the null hypothesis and there is a
signafacant difference in atleast one
passenger class
```

21. COVARIANCE & CORRELATION

↳ These concepts are heavily utilized in ML (Particularly Linear regression) to understand how to quantiative variables interact with one another.

→ Covariance

It mathematically tell us the direction in which two variables change together.

- **Positive covariance** : Both variables increases together.
  ex- as age increases, salary Also increases.

- **Negative covariance** : One variable increases while the other decreases.
  ex- as the price of a car increases, the Fuel usage increases.

↳ Formula

Deviation of x from its mean — Deviation of y from its mean

$$cov(x,y) = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

(n-1: Degree of freedom)

- $cov(x,y)$ - covariance between X & Y
- x_i = i^{th} value of variable x
- y_i = i^{th} value of variable y
- $\bar{x}$ = Mean (average) of X
- $\bar{y}$ = Mean (average) of Y
- n = no of observations (sample size)

- $cov > 0$ → Positive relationship
- $cov < 0$ → Negative relationship
- $cov = 0$ → No linear relationship

→ **Correlation (the strength)** : shows the strength & direction of the relationship.

↳ **Formula**

$$r = \frac{cov(X,Y)}{\sigma X \, \sigma Y}$$

- $cov(X,Y)$ = covariance between X & Y
- Range $-1 \leq r \leq 1$
- r = pearson correlation coefficient
- σX = standard deviation of variable X
- σY = standard deviation of variable Y

↳ **The scale** : correlation values will always fall between -1 & 1.

- $r = 1$ → Perfect positive correlation
- $r = -1$ → Perfect negative correlation
- $r = 0$ → No correlation.

↳ Machine Learning context

Feature with high correlation with the target are usually useful predictors, while features with near-zero correlation often contribute little to linear prediction models.